Baron and Kenny (1986)

The Foundational Regression-Based Approach

Why Mediation?

- Establishing that X affects Y is only the **beginning** of a causal inquiry
- The deeper question: **how** or **why** does the effect occur?
- A **mediator** M represents the mechanism through which X influences Y

Key distinction Baron & Kenny insisted on:

- **Moderator:** changes the *strength or direction* of $X \to Y$ — appears as an interaction term
- **Mediator:** explains the *mechanism* of $X \to Y$ — lies on the causal pathway

These were routinely conflated in psychology before 1986.

The Mediation Path Diagram

$$X \xrightarrow{a} M \xrightarrow{b} Y$$

$$X \xrightarrow{c} Y \quad \xrightarrow{\text{becomes}} \quad X \xrightarrow{c'} Y \text{ (controlling for } M)$$

- Path a: effect of X on the mediator M
- Path b: effect of M on Y, controlling for X
- Path c: total effect of X on Y
- Path c': **direct** effect of X on Y after controlling for M

Perfect mediation: $c' = 0$ — X has no effect on Y once M is controlled

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The Four-Step Procedure

To establish mediation, estimate three regression equations and verify:

1. **Step 1:** Regress M on X — confirm X significantly predicts M (path a)

2. **Step 2:** Regress Y on X alone — confirm X significantly predicts Y (path c)

3. **Step 3:** Regress Y on both X and M — confirm:

    - M significantly predicts Y (path b)
    - Coefficient on X is **smaller** than in Step 2

When all conditions hold, M is a mediator. The stronger the reduction in $c \to c'$, the stronger the mediation.

The Sobel Test

Sobel (1982) provides a significance test for the **indirect effect** ab:

$$\widehat{\text{se}}(ab) \approx \sqrt{b^2 s_a^2 + a^2 s_b^2}$$

where s_a, s_b are standard errors of paths a and b respectively.

- Allows a z-test of $H_0 : ab = 0$
- Became the standard compact test for mediation in applied psychology
- Used widely for two decades after the paper

What the Paper Gets Right

- Provided a **clear, testable, operational definition** of mediation
- Connected directly to regression tools researchers already knew
- Established the vocabulary — *moderator vs. mediator* — that the whole field now uses
- The four-step procedure is intuitive and easy to implement

The 1986 paper is one of the **most cited papers in all of social science** for good reason: it gave empirical researchers a concrete procedure where none existed before.

What the Paper Silently Assumes

1. Linearity and no interaction: OLS regression implicitly assumes X and M do not interact in producing Y — the direct and indirect effects are assumed to **add up** to the total effect.

2. No unmeasured $M \rightarrow Y$ confounding: The coefficient on M in Step 3 is treated as causal — requires no common causes of M and Y outside the model. **Never stated.**

3. No formal causal framework: No DAGs, no counterfactuals, no do-calculus. Coefficients are given causal meaning purely by convention.

4. No discussion of identifiability: Whether direct and indirect effects can even be recovered from data — even in a randomized trial — is never raised.

The Bridge to What Follows

These silent assumptions are precisely what the later literature attacks:

- **Robins & Greenland (1992):** show that stratifying on M is biased even under randomization — the nonidentifiability problem Baron & Kenny never asked
- **Pearl (2001):** provides formal counterfactual definitions — CDE and NDE — that give mediation estimands rigorous causal meaning
- **VanderWeele & Vansteelandt (2009):** make the no-interaction and no-confounding assumptions explicit as A1–A4
- **Andrews & Didelez (2020):** show cross-world independence can fail even without the confounding structures Baron & Kenny never considered

Baron & Kenny is a special case of all these frameworks — **valid only under strong parametric and no-confounding assumptions that must be explicitly defended.**